

Mini Game Hub Report

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1 Project Specification

Technologies Used:

- Bash
- Python 3.10
- Pygame
- NumPy
- Matplotlib

2 Overview

The Mini Game Hub is a secure multiplayer gaming platform where two authenticated users can play multiple games through a graphical interface. The system integrates Bash for authentication and Python for game execution, data recording, and analytics.

3 System Architecture

```
hub/  
|-- main.sh  
|-- game.py  
|-- leaderboard.sh  
|-- games/  
|   |-- tictactoe.py  
|   |-- othello.py  
|   |-- connect4.py  
|-- users.tsv  
|-- history.csv
```

Entry point:

```
bash main.sh
```

4 Authentication System

- Two users are authenticated using SHA-256 hashing.
- Passwords are securely stored (not plaintext).
- New users can register.
- After login, Python game engine is launched.

```
python3 game.py <username1> <username2>
```

5 Game Engine

The system uses a base class for all games:

- Player handling
- Turn switching
- NumPy-based board
- Win condition checking

6 Games Implemented

6.1 Tic Tac Toe

- 10×10 grid
- Win condition: 5 in a row

6.2 Othello

- 8×8 board
- Disc flipping mechanics

6.3 Connect Four

- 7×7 grid
- Gravity-based coin placement

7 Screenshots and UI

7.1 Game Menu

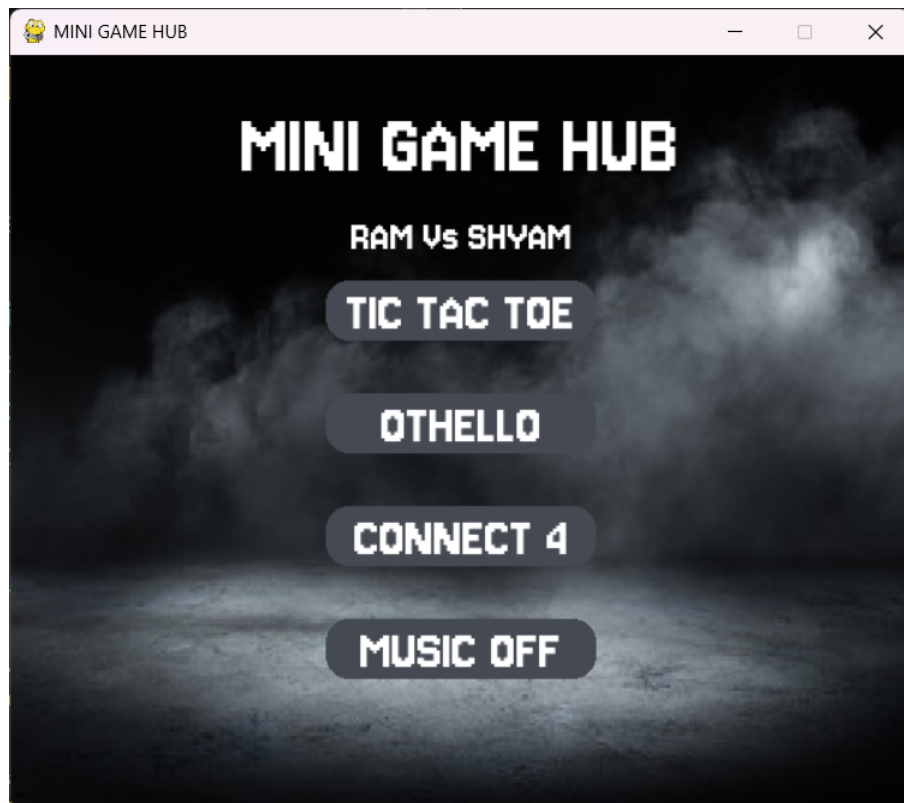


Figure 1: Main Game Menu

7.2 Tic Tac Toe

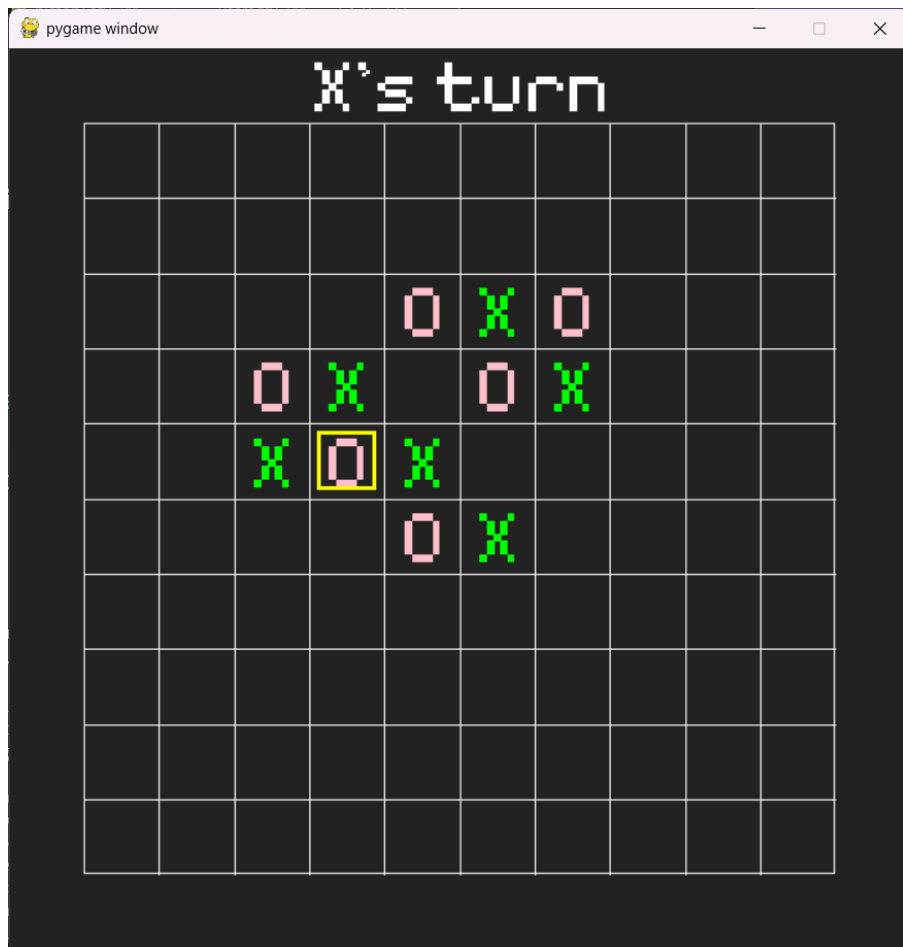


Figure 2: Tic Tac Toe Gameplay

7.3 Game Over

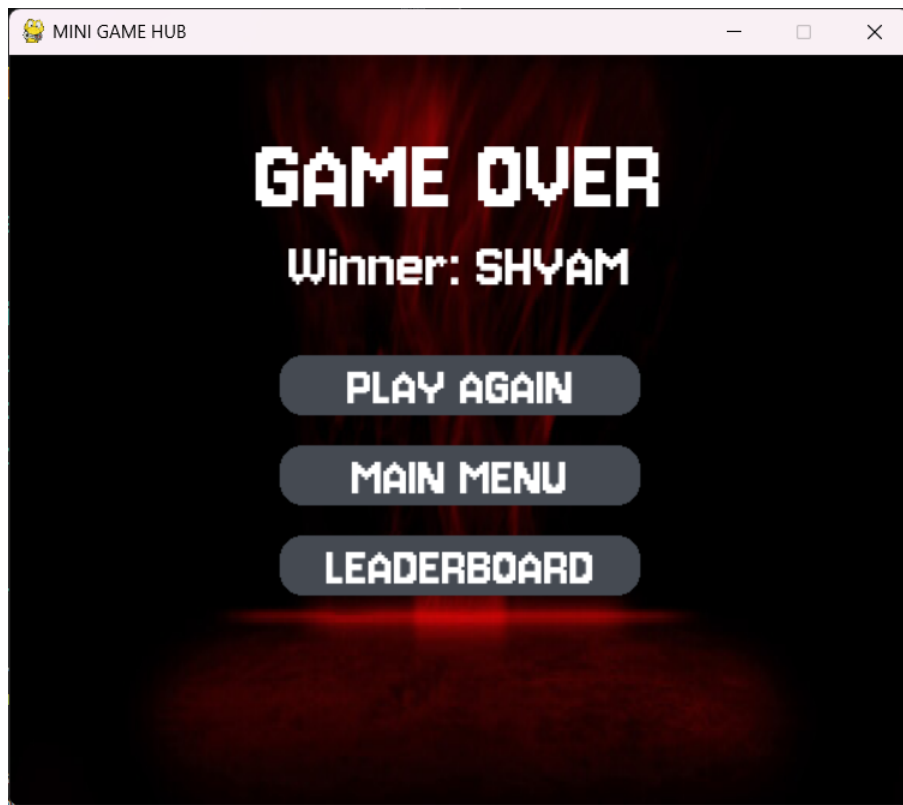


Figure 3: Game Over Screen

7.4 Leaderboard

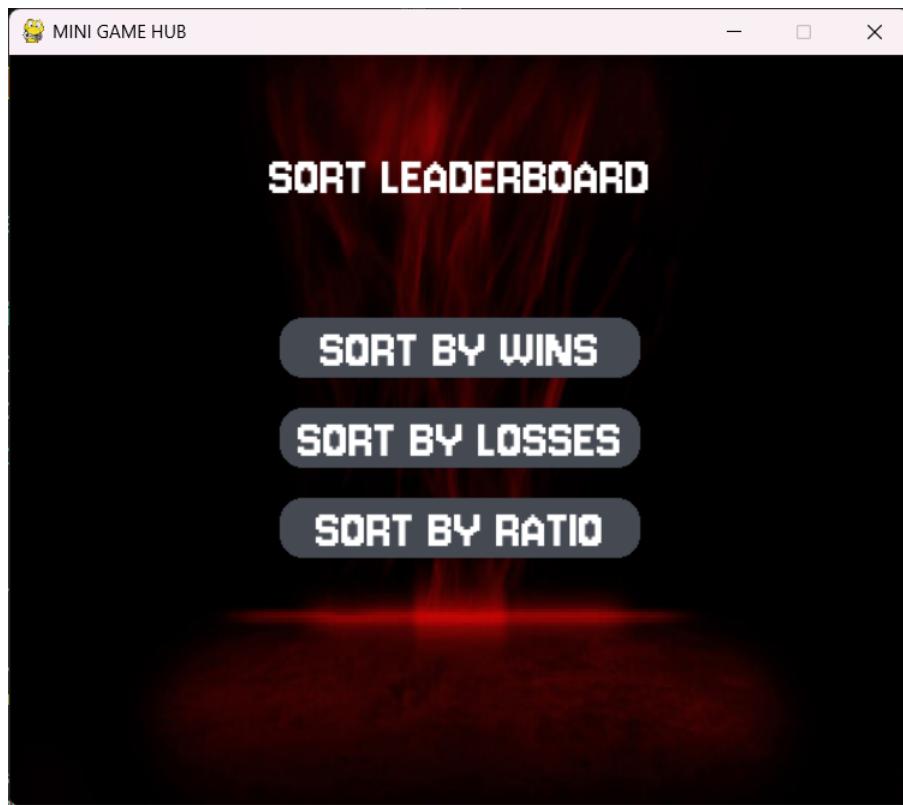


Figure 4: Leaderboard Interface

7.5 Graphs

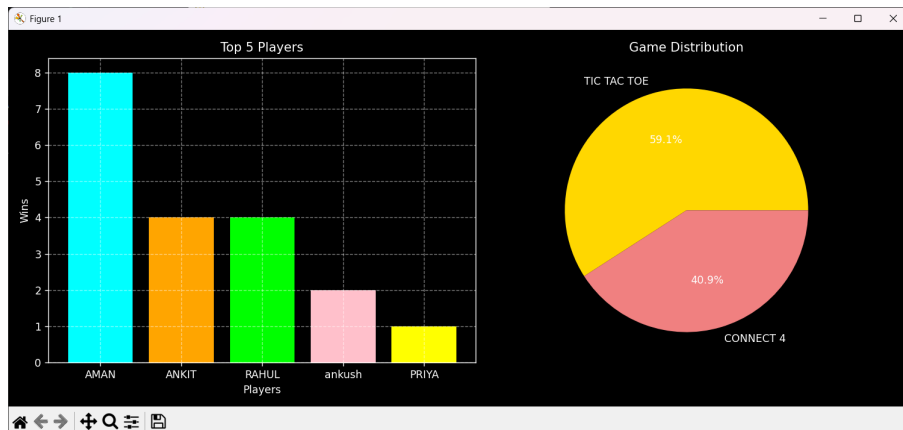


Figure 5: Statistical Graphs

8 Leaderboard and Analytics

- Game results are stored in history.csv
- Leaderboard is generated using Bash
- Displays wins, losses, and ratios

9 Technical Constraints

- NumPy used for all board representations
- Win logic implemented using vectorized operations
- No manual loops for win checking
- Full GUI using Pygame

10 Hurdles Faced

- Integrating Bash with Python
- Handling real-time UI in Pygame
- Implementing NumPy-based logic
- Managing CSV data consistency

Solutions:

- Used subprocess for communication
- Structured event loops properly
- Used vectorized NumPy operations

11 Future Improvements

- AI opponents
- Database instead of CSV
- Better UI animations
- Player ranking system

12 References

- <https://pyga.me>
- <https://numpy.org>
- <https://worldtohello.org>
- <https://boardgamegeek.com>